CSM
Fall 2025

EECS 16B
Worksheet 8

1. RLC Bandstop Filter

One way to compose a bandpass filter is by combining a low pass and high pass filter via a buffer. Another way to compose a bandpass is to use a RLC circuit of the following form:

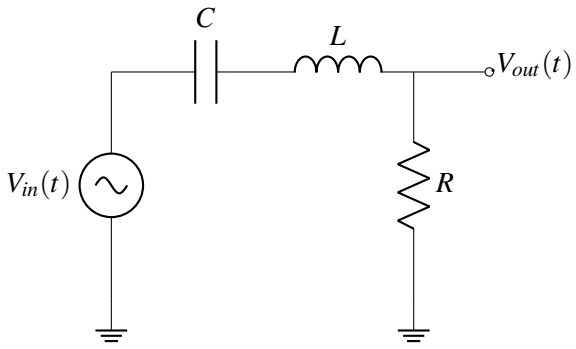


Figure 1: RLC Bandpass Filter

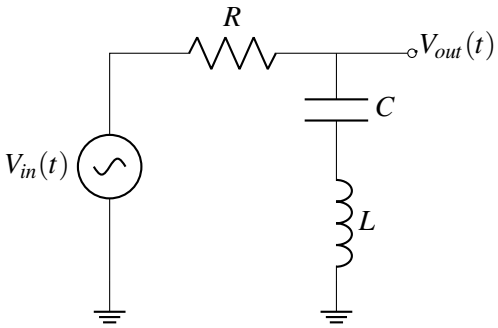


Figure 2: Unknown Behavior

Let's explore what happens when swap the location of the resistor with the location of the capacitor and inductor.

(a) Write out the transfer function of the circuit of unknown behavior.

(b) What is the magnitude of the transfer function?

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(c) Sketch the magnitude of this transfer function.
HINT: The name of this question is bandstop filter. What do you think that means?

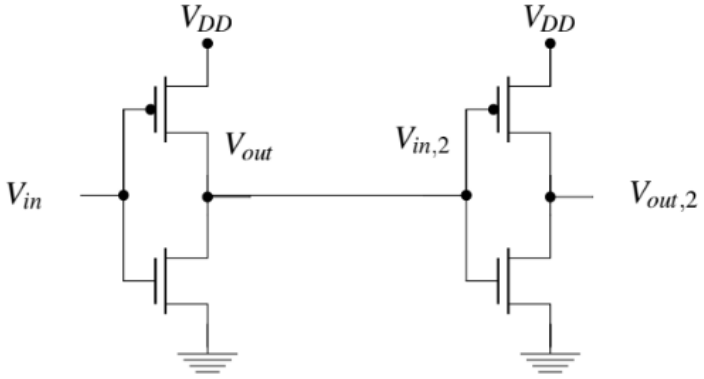
(d) When is the magnitude of the transfer function zero? When is it one?

(e) How would you describe the behavior of this circuit? Why do you think circuits with this type of circuit behavior are classified as notch/bandstop filters? Why is this specific filter called a *resonsant* bandstop filter?

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2. Transistor Switch Model

You have two CMOS inverters made from NMOS and PMOS devices. Both NMOS and PMOS devices have an "on resistance" of $R_{on} = 1\,\text{k}\Omega$, and each has a gate capacitance (input capacitance) of $C = 1\,\text{fF}$ (femto-Farads = 10^{-15}). We assume the "off resistance" (the resistance when the transistor is off) is infinite (i.e., the transistor acts as an open circuit when off). The supply voltage V_{DD} is 1 V. The two inverters are connected in series, with the output of the first inverter driving the input of the second inverter.



(a) Assume the input to the first inverter has been low ($V_{in} = 0\,\text{V}$) for a long time, and then switches at time $t = 0$ to high ($V_{in} = V_{DD}$). **Draw a simple RC circuit and write a differential equation describing the output voltage of the first inverter for time** $t \geq 0$. Don't forget that the second inverter is "loading" the output of the first inverter - you need to think about both of them.

(b) **Sketch the output voltage of the first inverter, showing clearly (1) the initial value, (2) the initial slope, (3) the asymptotic value, and (4) the time that it takes for the voltage to decay to roughly $\frac{1}{3}$ of its initial value.**

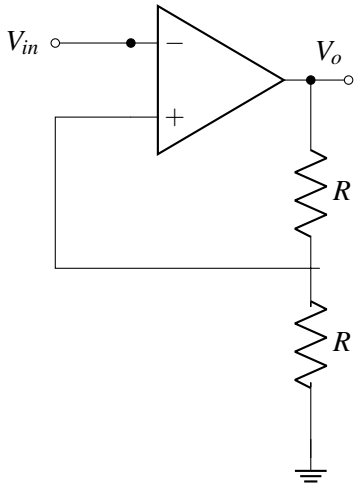
(c) A long time later, the input to the first inverter switches low again. **Sketch the output voltage of the first inverter, showing clearly (1) the initial value, (2) the initial slope, and (3) the asymptotic value.**

(d) For each complete input cycle described in the two steps above, **how much charge is pulled out of the power supply?** Give both a symbolic and numerical answer.

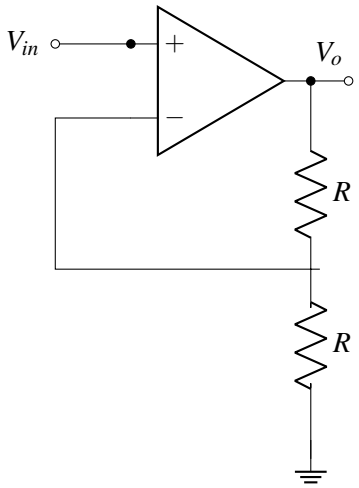
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3. OpAmps as Amplifiers

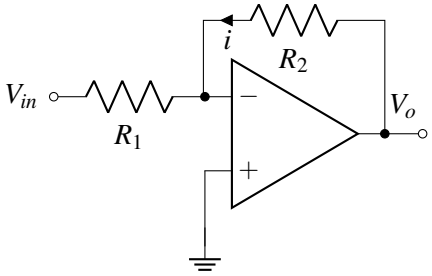
(a) In general, what is feedback? What are positive and negative feedback? And how do you introduce feedback into op-amps?



(b) Referring to the above example of an op-amp employing positive feedback, start with the assumption that $V_+ = V_-$. Now, taking into account the fact that $V_{out} = A(V_+ - V_-)$ (where A is some gain), try to raise V_{out} (by messing with V_{in}) and observe what happens. Try to lower V_{out} and observe that too.

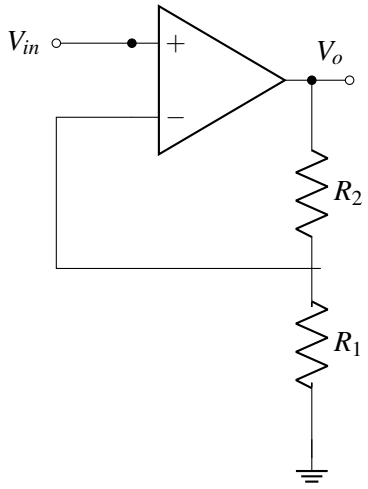


(c) Now do the same with this above example of an op-amp that uses negative feedback



(d) In terms of the input V_{in}, what is V_{out} for this circuit? What is the gain of this circuit?

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(e) Again, in terms of the input V_{in}, what is V_{out} for this circuit? What is the gain?

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